This minimalist approach has two aspects. First, we take the wave function seriously as a direct representation of reality, not just a bookkeeping device to help us organize our knowledge. We treat it as ontological, not epistemic. That's the most austere strategy we can imagine adopting, since anything else would posit additional structure over and above the wave function. But it's also a dramatic step, since wave functions are very different from what we observe when we look at the world. We don't *see* wave functions; we see measurement outcomes, like the position of a particle. But the theory seems to demand that wave functions play a central role, so let's see how far we can get by imagining that reality is exactly described by a quantum wave function.

Second, if the wave function usually evolves smoothly in accordance with the Schrödinger equation, let's suppose that's what it always does. In other words, let's erase all of those extra rules about measurement in the quantum recipe entirely, and bring things back to the stark simplicity of the classical paradigm: there is a wave function, and it evolves according to a deterministic rule, and that's all there is to say. We might call this proposal "austere quantum mechanics," or AQM for short. It stands in contrast with textbook quantum mechanics, where we appeal to collapsing wave functions and try to avoid talking about the fundamental nature of reality altogether.

A bold strategy. But there's an immediate problem with it: it sure *seems* like wave functions collapse. When we make measurements of a quantum system with a spread-out wave function, we get a specific answer. Even if we think an electron wave function is a diffuse cloud centered on the nucleus, when we actually look at it we don't see such a cloud, we see a point-like particle at some particular location. And if we look immediately again, we see the electron in basically the same location. There's a good reason why the pioneers of quantum mechanics invented the idea of wave functions collapsing—because that's what they appear to do.

But maybe that's too quick. Let's turn the question around. Rather than starting with what we see and trying to invent a theory to explain it, let's start with austere quantum mechanics (wave functions evolving smoothly, that's it), and ask what people in a world described by that theory would actually experience.

Think about what this could mean. In the last chapter, we were careful to talk about the wave function as a kind of mathematical black box from